

FANTASY OR SCIENCE?

the golden age of alchemy

The story of alchemy is one of magic versus science, of secrecy and symbolism, and of seeking perfect understanding in order to achieve transformation. Alchemists explored the nature of matter; pursued a holy grail of turning base metals, such as lead, into gold; and sought perfection of the body and soul.

The Renaissance was a golden age for alchemy. Its practitioners built on traditions said to have begun in the ancient world—including links between alchemy and Hermeticism (see pp.134–135, 150–151), both of which claimed Hermes as a founding figure—and added new layers of knowledge and symbolism. Their work contributed to the split between early modern science and more esoteric material, which also occurred during this period.

Alchemy and chemistry

Renaissance alchemy also had connections with the alchemy of Arabic practitioners, such as Geber (see p.150). Arabic influence might be seen in the terms “chemistry” and “alchemy,” which were often used interchangeably during the Renaissance: although the root of “chem,” found in both, is much disputed, it may be derived from the Arabic word *kimiya* (transmutation).

Many Renaissance alchemists carried out what we would now consider scientific research, laying the foundations for later laboratory work in areas such as chemistry and medicine. However, at the time, many were considered charlatans, duping people with promises of riches. By the early 1700s, modern science was emerging, and “alchemist” was the name increasingly given to those chasing what began to be perceived as unscientific magical goals—such as the creation of gold. The work of Belgian scholar Jan Baptista van Helmont is often said to represent the shift from alchemy to science: he believed in alchemical principles, but also held that knowledge of nature could only be gained by experimentation.

◀ Precursor to chemistry

This image of a double still comes from German surgeon and alchemist Hieronymus Brunschwig’s 16th-century distillation manual. A pioneering book on chemistry, its ideas borrowed heavily from alchemy.



All things must change

Transmutation and purification—turning something base into something better—were concepts central to alchemy, as were attempts to create the elusive Philosopher’s Stone, which were shrouded in secrecy. The Stone was believed to enable great feats: not only the ability to turn base metals into precious ones, but also the curing of illness, the attainment of immortality (through the related Universal Panacea and Elixir of Life), and the purification of the soul. Behind such beliefs lay the ancient idea that all things naturally strive to achieve perfection.

Secrecy and symbols

In order to achieve their aims, alchemists followed a secretive multi-staged process, much of it laden with symbolism linked to religion, natural philosophy, and popular magic. It involved ritualistic practices and sequences of numbers, but also procedures that we would recognize from modern science, such as distillation (see left).



▲ Fixing the volatile

This 1600s or 1700s illustration is based on the earlier, unillustrated text of the *Twelve Keys of Basil Valentine* (1599), a work that details 12 steps to creating the Philosopher’s Stone. The flowers may symbolize purified precious metals.

▼ Pursuing the Stone

This “Ripley Scroll” is named after the influential 15th-century English alchemist George Ripley. The red and green lions represent sulfur and mercury respectively: two central ingredients for making the Stone.